Demographic Biases in Naturalistic Language Recordings in the CHILDES Database

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Abstract

In recent years, the importance of estimating demographic biases in research has become apparent. Here we provide a systematic review of the CHILDES database, the major source of naturalistic recordings of children's linguistic environment. We analyzed the database according to four dimensions considered central to language learning: SES, urbanization, family structure, and language. We present descriptive statistics of each dimension to assess whether naturalistic recordings were biased regarding the demographics of the countries and the families recorded within them. We find that CHILDES's recordings overrepresented wealthier countries and higher parental education levels; urban settings; and smaller households. Middle- and higher-class participants were likewise over-represented. The corpora were not representative of their countries in terms of urbanization either - with a larger percentage of families residing in urban settings than is overall true for their respective countries. In terms of family structure, nuclear families were more prevalent than in the countries where the data was collected. Last, we found that corpora were linguistically diverse, but we estimate that these recordings underrepresented bilingual and multilingual households. We conclude that researchers should be mindful when generalizing from naturalistic recordings of children's input and output obtained from CHILDES, and make recommendations for the future of the use of CHILDES.

Keywords: childes, verbal input, infant-directed speech, language, weird

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Research highlights

- We examined CHILDES, the major source of data on children's linguistic environment, for potential sampling bias in terms of SES, urbanization, family structure, and language.
- We found that the 48 countries represented in CHILDES overrepresented higher-GDP, educated, urbanized countries with small nuclear families.
- Within these countries, corpora overrepresented higher-SES, educated, urban, nuclear, and monolingual families.
- Interpretation of studies using CHILDES should acknowledge these biases.

Since its foundation in 1984, CHILDES (the Child Language Data Exchange System, MacWhinney, 2000) has been the major source of naturalistic recordings and transcript data for researchers studying language acquisition. Moreover, it has contributed to establishing seminal work in various domains such as Theory of Mind (Bartsch & Wellman, 1995) and memory (Anderson & Schooler, 1991). It has also inspired theories that help understand the connection between language input and development (Christiansen, Allen, & Seidenberg, 1998). As such, CHILDES has been and is one of the most valuable assets of our field. While much of previous work relied on one or a handful of corpora, even when all relevant corpora are used (Ibbotson & Browne, 2024; Tan, Loukatou, Braginsky, Mankewitz, & Frank, 2024), conclusions are more likely to be generalizable if they are based on representative data. We examine the extent to which CHILDES' naturalistic recordings, containing information about both input and output, come from a representative sample of the world's populations.

Researchers in many fields are increasingly aware of the bias their samples may have towards WEIRD populations (Cychosz & Cristia, 2022; i.e., Western, Educated, Industrial, Rich, Democratic, Henrich, Heine, & Norenzayan, 2010; Moriguchi, 2022; Nielsen, Haun, K??rtner, & Legare, 2017; Singh, Cristia, Karasik, Rajendra, & Oakes, 2023). Recent calls have been made to diversify research in psychology and developmental science to address this issue (Blasi, Henrich, Adamou, Kemmerer, & Majid, 2022; Kidd & Garcia, 2022; Majid & Levinson, 2010). For instance, Kidd and Garcia (2022) systematically reviewed publications from child-language journals. They revealed biases towards specific continents (North America and Europe) and languages (English, Spanish, French), highlighting the need for increased diversity in the populations and languages studied.

Every area of research conceptualizes different dimensions relevant to explaining variance for a given research topic. In what follows we present the four most relevant dimensions thought to play a role in language acquisition, particularly related to naturalistic recordings: socioeconomic status, urbanization, family structure, and language.

Socioeconomic Status. It is beyond the scope of this paper to detail all the theories that attempt to account for the complex causal pathways that may connect socio-economic status (SES) to children's language environments Golinkoff, Hoff, Rowe, Tamis-LeMonda, & Hirsh-Pasek (2019). While we do not take a stance on whether, how, or why SES and language environments relate to each other, SES is one of the factors that has been repeatedly studied in the context of early language acquisition, and as such, deserves attention. For example, Hoff (2003) compared the speech of high- versus middle-SES American mothers. College-educated mothers produced more utterances to their children, with more diverse vocabulary, longer phrases, and a higher number of utterances continuing a topic the child had brought up, which some claim to also be related to language outcomes [see also Hart and Risley (1995);Hoff-Ginsberg (1990);Hoff (2014); Huttenlocher, Vasilyeva, Cymerman, and Levine (2002); Huttenlocher, Vasilyeva, Waterfall, Vevea, and Hedges (2007); see also Dailey and Bergelson (2022); Leonardo, Havron, and Cristia (2022); for meta-analyses supporting the link; and Bergelson et al. (2023) for a large-scale study finding no SES effects].

In the developmental literature, SES is primarily indexed by parents' education (Ensminger & Fothergill, 2014; Hoff, 2003), but can also be indexed by parental income, occupation, or a composite measure of these three (e.g. using the Hollingshead index, Hollingshead et al., 1975).

Urbanization. This dimension represents differences across populations in terms of the extent to which they are organized around urban sites, which often entail differences in job opportunities, living conditions, and infrastructure. There have been proposals that societies varying in their urbanization have differing views and values about the role of children in conversations (Draper & Harpending, 2017; Keller, 2012; Richman, Miller, & LeVine, 1992; e.g., Sharma & LeVine, 1998). For instance, Keller (2012) discusses three prototypical cases, urban, rural, and hybrid, which differ in terms of their goals for children: Urban families aim for child psychological independence, rural families for child physical autonomy and interdependence, and hybrid families for some mix. Vogt, Masson-Carro, and Jong (n.d.) found that the number of gestures, gesture-speech alignment, and gesture types all vary across the three groups in ways that can be related to Keller's typology, in urban Dutch, urban Mozambique, and rural Mozambique. Similarly, Cristia (2023) systematic review uses urbanization to describe the frequency of child-directed vocalization, finding that it is far less frequent in rural than urban communities.

Family structure. In this paper, we use the concept of family structure to encompass a variety of related aspects such as birth order and the way that caregiving is structured, each of which has an impact on child and language development (e.g., Blake, 1981; Bornstein, Putnick, & Suwalsky, 2019; Duncan & Paradis, 2020; Havron, Lovcevic, et al., 2022; Havron et al., 2019; Hoff-Ginsberg & Krueger, 1991; Tomasello & Mannle, 1985). For instance, children with older siblings exhibit lower language skills compared to first-born children across various cultures (e.g., Peyre et al., 2016 in France; Havron, Lovcevic, et al., 2022 for Singapore; Zambrana, Ystrom, & Pons, 2012 for Norway), although second-born children might benefit from overhearing conversations between their caregivers and older sibling (Oshima-Takane, Goodz, & Derevensky, 1996). As for caregiving structures, in many middle-class Euro-American families, parents typically assume primary responsibility for children, with the mother as the primary caregiver (e.g., Bianchi & Milkie, 2010). However, certain cultures, like the Turkish families described by Isleyen (2021), adopt a different approach, with nuclear families living in separate apartments but sharing common spaces and caregiving responsibilities, resulting in extensive support networks.

Languages. Linguistic factors likely influence the development of language skills in children (Blasi et al., 2022; Kidd & Garcia, 2022). For example, based on transcriptions of conversations, Pye and colleagues documented that K'iche' Mayan children frequently use and understand passive constructions from a very young age, unlike their English-speaking peers, refuting the idea that passive constructions can only emerge later in development

(C. L. Pye, 1980; C. Pye & Poz, 1988). Similarly, many Indo-European languages show a strong noun bias in early vocabularies (a bias for acquiring words for concrete referential objects rather than actions), which has been claimed to be a universal feature of early language acquisition. However, studies have shown that in some Mayan languages, including Tseltal and Tsotsil (Casillas, Foushee, Méndez Girón, Polian, & Brown, 2024; De León, 1999), there is a verb bias instead. Thus, generalizations about language acquisition could be inaccurate when they rely on a linguistically biased database.

Another major dimension is the study of bilingualism or multilingualism McCabe et al. (2013). There is evidence that monolingual and bilingual early language development differ in some aspects, particularly regarding phonological acquisition and word learning. For example, monolingual infants' ability to discriminate non-native sounds declines during the first year of age, whereas infants exposed to more language(s) maintain the discrimination window for a longer period (Singh, Rajendra, & Mazuka, 2022). In terms of input, bilingual linguistic exposure is divided between two or more native languages. It has been shown that the amount of exposure to each native language can affect bilingual infants' speech discrimination abilities (Garcia-Sierra et al., 2011).

The current study. Inspired by calls to document the demographic and linguistic diversity of participant samples (Kidd & Garcia, 2022; Singh, Barokova, et al., 2023), we sought to provide a systematic analysis of the naturalistic speech corpora in the CHILDES database by quantifying the diversity of each dimension introduced above (SES, urbanization, family structure, and languages). Following Ghai (2021) recommendation to enhance the description of diversity in behavioral sciences, we examined different levels, from a macro-level, involving broad country comparisons, to a corpus-level, where we analyzed individual corpora for a more detailed and nuanced understanding of the data.

Methods

Inclusion criteria. We extracted information from the online CHILDES database in 2021, at which point there were 430 corpora. We excluded corpora on clinical populations (N = 69), in-lab recordings (N = 26), and elicitation tasks (N = 114), because they would have introduced additional sources of variation and biases. For example, the inclusion of clinical populations may highlight differences in access to diagnosis in different countries. Given our interest in naturalistic input and output represented in CHILDES, we also excluded corpora where only child speech was transcribed (N = 19) and diary studies (N = 11). Finally, we excluded 11 corpora that were not available, resulting in a final sample of 180 corpora containing naturalistic recordings of children's input and output.

Demographic Data Collection. Data on participant demographics was extracted from the references provided for each corpus in CHILDES such as articles, book chapters, dissertations, and the actual content of transcribed recordings (for more information please see SM1; Anonymized, 2024). When key variables could not be determined from these sources, we contacted corpus curators for missing information, and over a third (103 corpora, 32%) provided additional data or confirmed precoded information. They were also invited to correct our coding if it was inaccurate to their knowledge, and we took their coding as definitive. Unfortunately, curators for 50 corpora (15%) could not be contacted due to unresponsive email addresses (38 corpora), or the curator's passing (12 corpora).

Analyses. Analyses and visualizations were carried out using R (version 4.4.2, R Core Team, 2020) and ggplot2 (Wickham, 2016). Data, scripts, and Supplementary Materials are available on the (OSF)[https://osf.io/q9w82/].

Results

Do Countries in Our CHILDES Sample Represent the World's Countries?

The 180 corpora included in this study represent 48 different countries or territories across

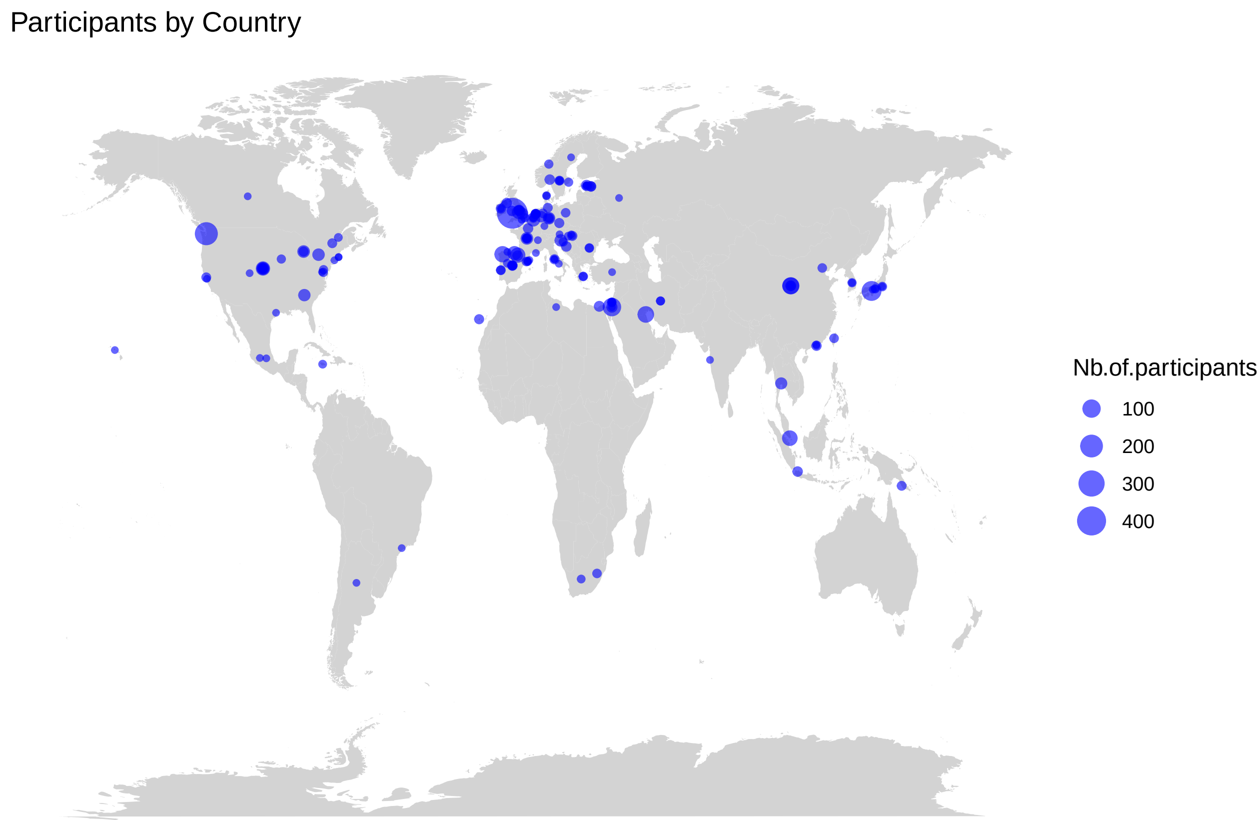

Figure 1. Geographical Distribution of the Corpora in CHILDES. Each circle represents a corpus, and its size is proportional to the number of participants.

all populated continents (see Figure 1). 28 countries, representing 151 corpora (84%), belong to the OECD (Organisation for Economic Co-operation and Development). Meanwhile, OECD countries represent only 19.5% of the world's countries. The country with the most corpora is the United States with 30 different corpora, followed by Spain with 24, followed by the United Kingdom and France with 11.

Analysis of the number of utterances per country presents a different picture, with the United Kingdom having the largest number of utterances, followed by the United States, Japan, Indonesia and Spain (see Figure 2).

We first assessed the representativeness of our CHILDES sample by comparing country-level descriptors across - SES (education and GDP), urbanization, and family

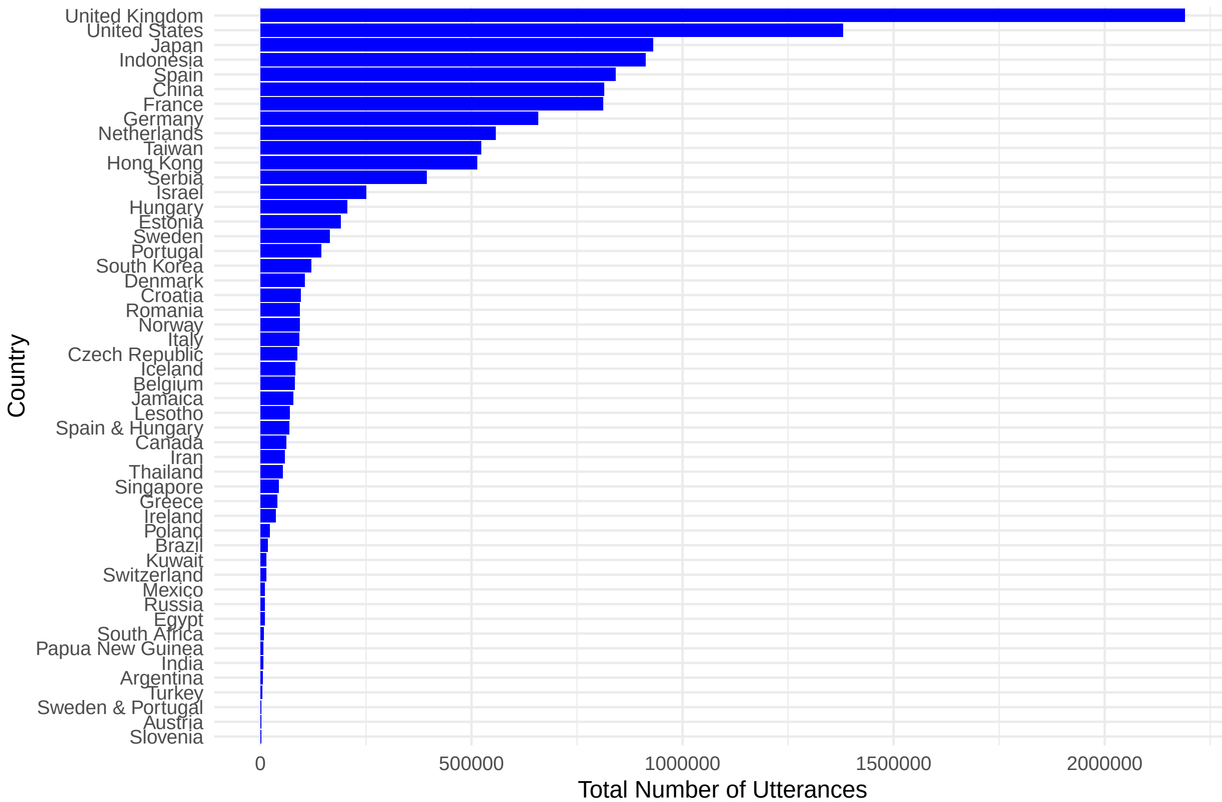

Figure 2. Number of utterances per country in CHILDES.

structure - against global distributions. We used public datasets from Our World in Data, the World Bank (World Development Indicators, WDI, Arel-Bundock, 2021), and the UN Household Size and Composition database (United Nations, 2022, see full details in SM1) to extract these global descriptors.

The countries in our sample of CHILDES differed significantly from the distribution of countries in the world on our key descriptors according to unpaired tests that did not assume equality of variances (Welch's t): a higher proportion of the population residing in our sample's countries completed lower secondary school than the worldwide sample (% compl. LSS, t(130.41)=-6.19; the countries were more urban (% urban, t(79.48)=-3.44, $p < .001$) and richer (log GDP per capita, t(102.35)=-6.02, $p < .001$); and they had smaller households (average household size, t(118.80)=7.12,$p < .001$; see Figure 3).

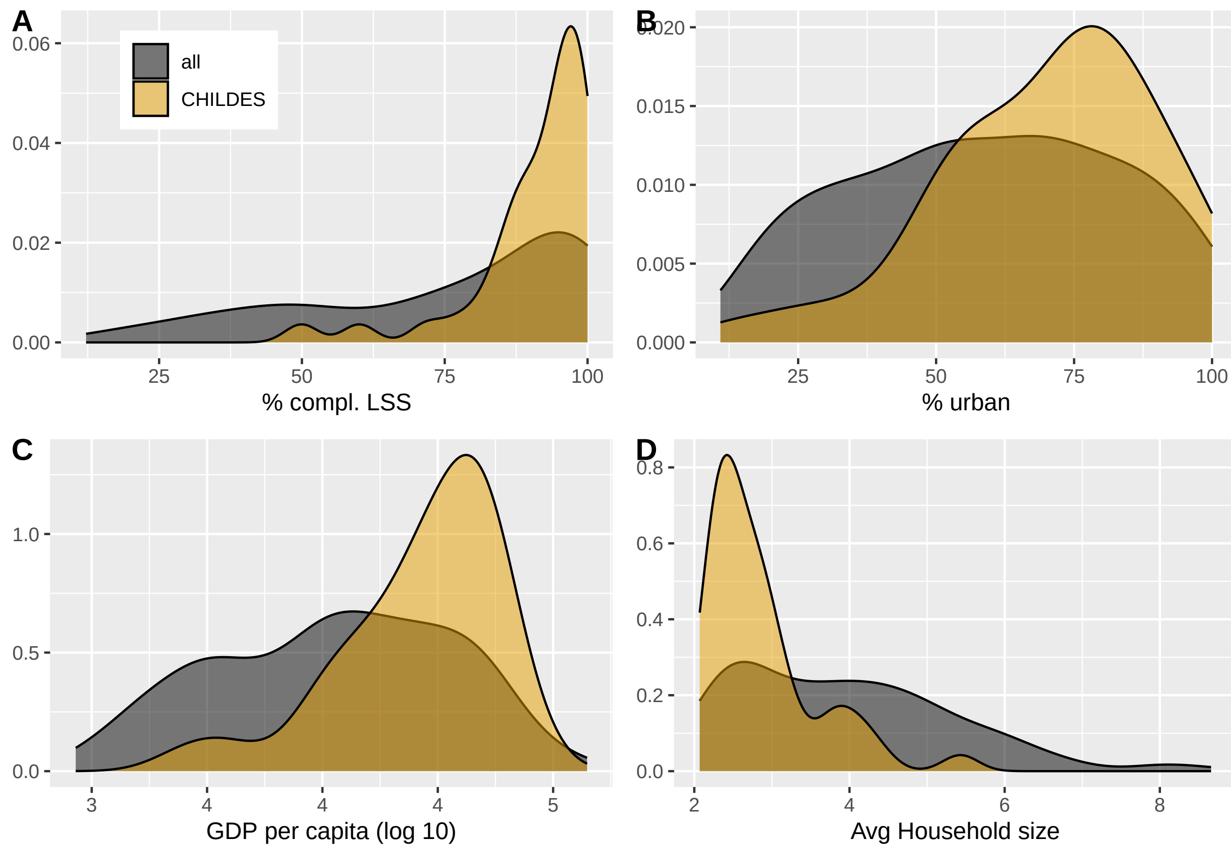

Figure 3. Comparison of Country-Level Descriptors Between World Countries and Our CHILDES Sample. Note. Density plots show the distribution of country-level descriptors for all world countries (gray) versus countries represented in our sample of CHILDES (orange). % compl. LSS is the percentage of the country's population completing lower secondary school. % urban is the percent of the country's population residing in urban (as opposed to rural) locations. GDP stands for Gross Domestic Product. Avg Household size stands for average household size. To clarify, the calculations are based on unweighted country-level statistics (each country contributes equally to the distribution irrespective of population size). For example, panel A indicates that, worldwide, only a minority of countries have over 90% completion of lower secondary school, whereas in the CHILDES sample, the density curve shows a higher concentration of countries in this range.

Table 1

Variables of interest

Dimension	Corpus.level.variable	
SES	Parents' socio-economic status	Low, mid and/or high
SES	Parents' education level	Highest level of education completed: Primary, Hig
SES	Parents' occupation	Parents' activity or profession
Urbanization	Type of community	Urban, rural, mixed
Family structure	Household composition	Whether the family was composed primarily of care
Family structure	Percent children with sibling(s)	The percentage of children in the corpus who had a
Family structure	Average number of siblings	How many siblings children had on average (includi
Language	Language(s) spoken	Which languages were spoken in the transcripts and
Language	Lingual status	Whether more than one language is spoken in the c

How Representative and Diverse are Participant Samples? We next investigated variables characterizing the specific families within each corpus, as far as possible given substantial missing data (see Table 1 and figures below for definitions of the variables and the number of corpora represented in each analysis, see SM1 for more details on variables).

Socio-economic Status: SES, Education, and Occupation. The vast majority of families were of middle or higher SES. In about a fifth of the corpora samples were described as mixed (spanning lower to middle or higher SES). A very small percentage were described as families of low SES. As we show in SM2, this general distribution over-represents higher SES families. A smaller proportion of corpora (additionally) reported parental education level (Figure 4). Among families in CHILDES, three-quarters included at least some children whose parents had a graduate or postgraduate degree. In fact, over half of the corpora contained data solely from children whose parents had at least a graduate degree. Overall, more educated parents were overrepresented when compared to their source countries, which is reinforced by the observations that parents had professions linked to graduate-level education in over half of corpora (see SM3).

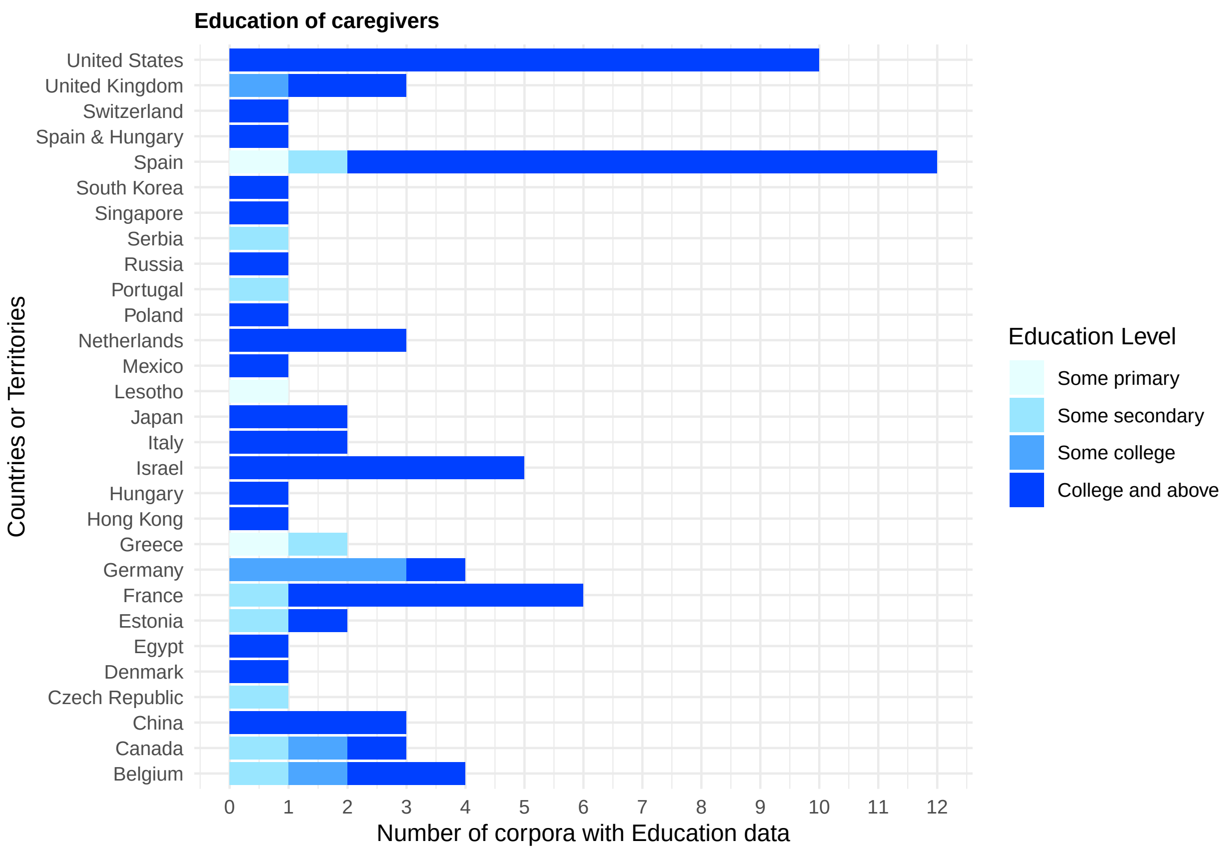

Figure 4. Distribution of Minimum Levels of Parental Education by Country. Spain & Hungary corresponds to a corpus collected in two countries. The order of countries is reversed-alphabetical. Note that the figure displays data only from countries for which we have available information. Corpora with missing data are not represented in the figure The same applies for Figures 5-8.

Urbanization. The majority of corpora were urban (Figure 5), with an over-representation of urban households given the source countries (see SM4).

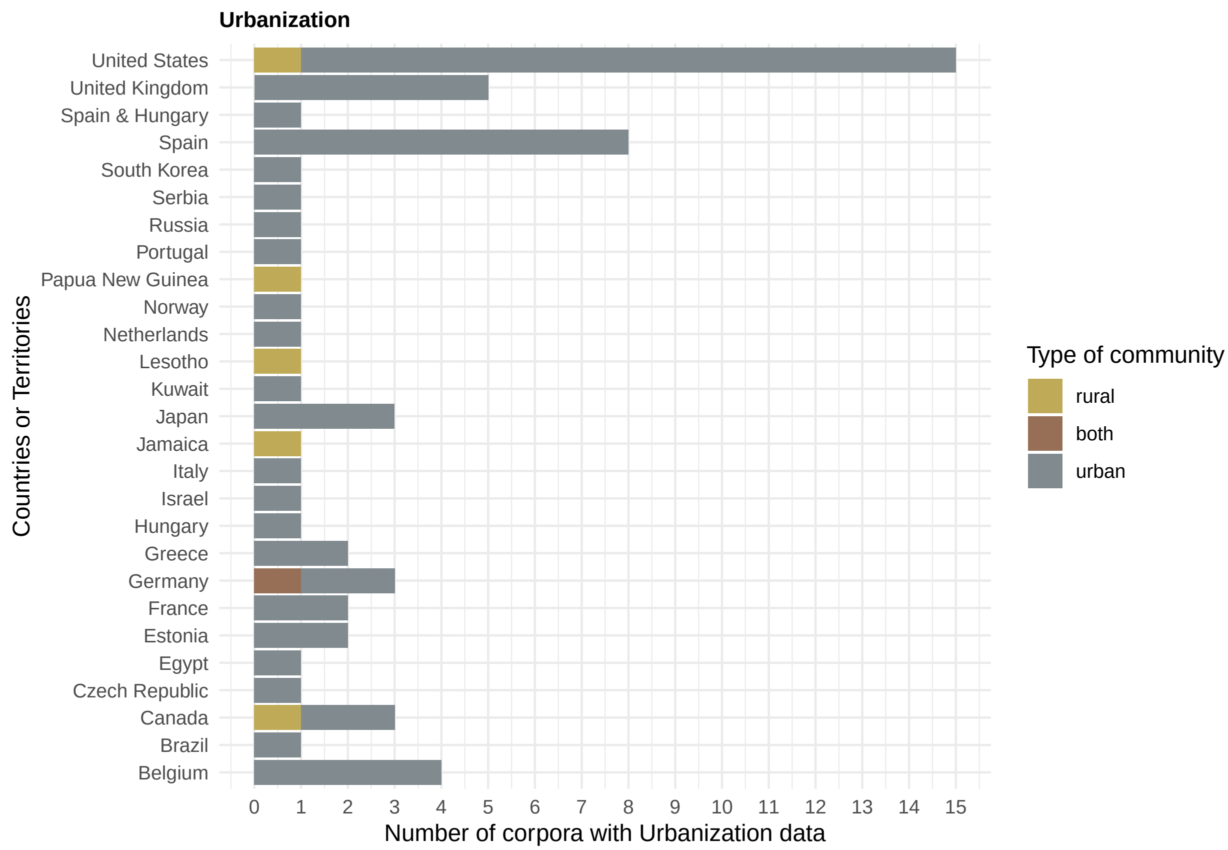

Figure 5. Distribution of Type of Community by Country.

Family Structure: Household composition and number of siblings. As Figure 6 shows, the overwhelming majority of corpora contained only nuclear families (i.e., parents and their children), and only a tiny percentage had a mixture of nuclear and extended families. We were not able to find a systematic database allowing us to check whether this was representative of the source countries.

Next, about a quarter of the corpora in our CHILDES sample consist exclusively of singletons (children with no siblings), while the remaining include children who have at least one sibling (Figure 7; see SM5 for why singletons might not be underrepresented).

Languages. We found a total of 63 different languages or language combinations (for bilingual and multilingual children; see SM6). About a third of the included corpora were not monolingual. According to information in SM7, our sample of CHILDES

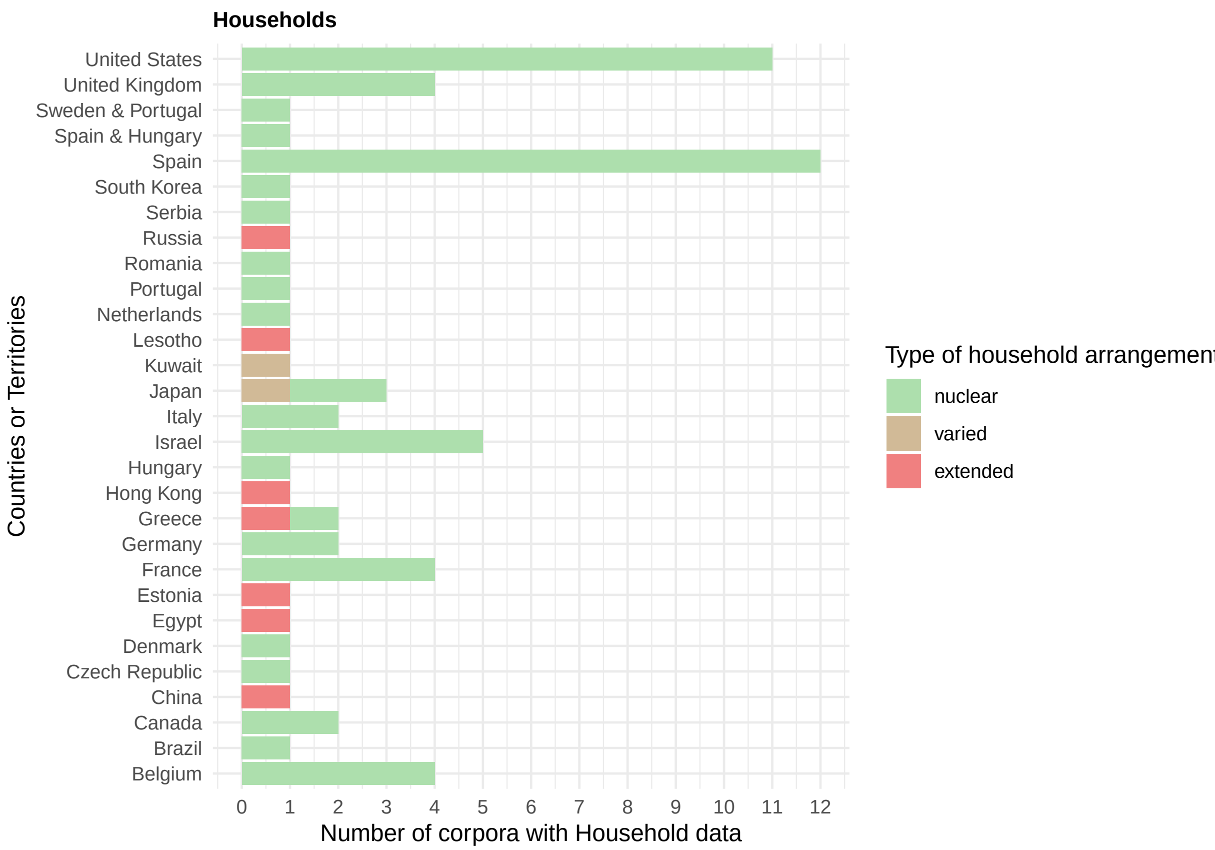

Figure 6. Distribution of Household Arrangement by Country. Sweden & Portugal corresponds to a corpus collected in two countries. The same applies for Figure 8.

underrepresents bilingual and multilingual households in the world. As for linguistic diversity, most of the corpora pertain to Indo-European languages (Figure 8). English is the most prevalent language, with 50 corpora (30 exclusively monolingual), comprising 28% of the included corpora. Additionally, it is the language most studied for bilingualism or multilingualism, with English being combined with: Cantonese, Danish, Dutch, Farsi, French, Hebrew, Hungarian, Japanese, Mandarin, Portuguese, Russian, Spanish, and Swedish. The next most represented language in terms of different corpora is Spanish, with 25 corpora (14 monolingual).

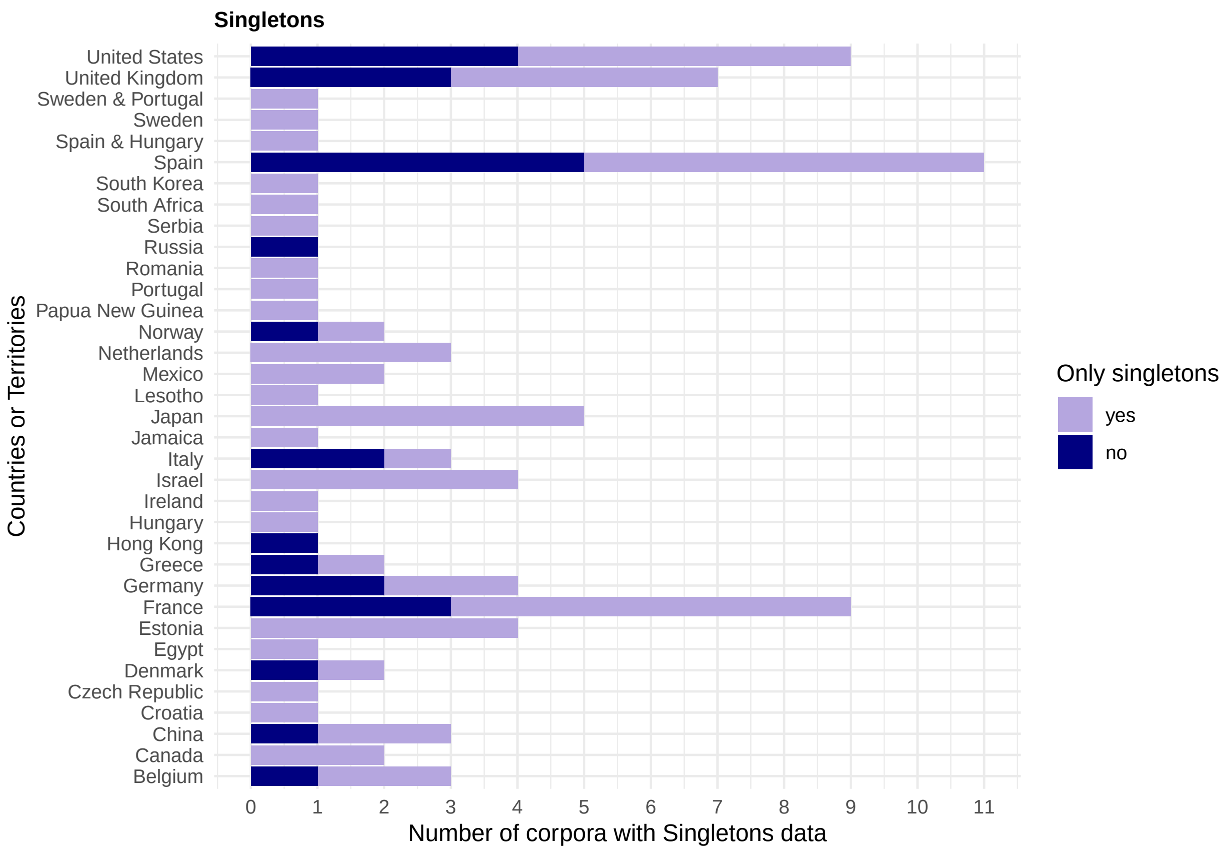

Figure 7. Distribution of Singletons by Country.

Discussion

The current paper provides a systematic review of corpora of naturalistic recordings in the CHILDES database, a priceless public research resource. We assessed and characterized sampling bias by examining the database through four dimensions believed to be relevant for early language acquisition: SES, urbanization, family structure, and languages.

Thanks to the collective work of nearly 300 individuals (with deep thanks to the people acknowledged in SM8), we included 180 corpora, in 63 different languages or language combinations, from 48 different countries, primarily OECD (84%). Despite a remarkable diversity in terms of languages and countries, analyses of the demographic characteristics of our CHILDES sample revealed several measures where the corpora do not

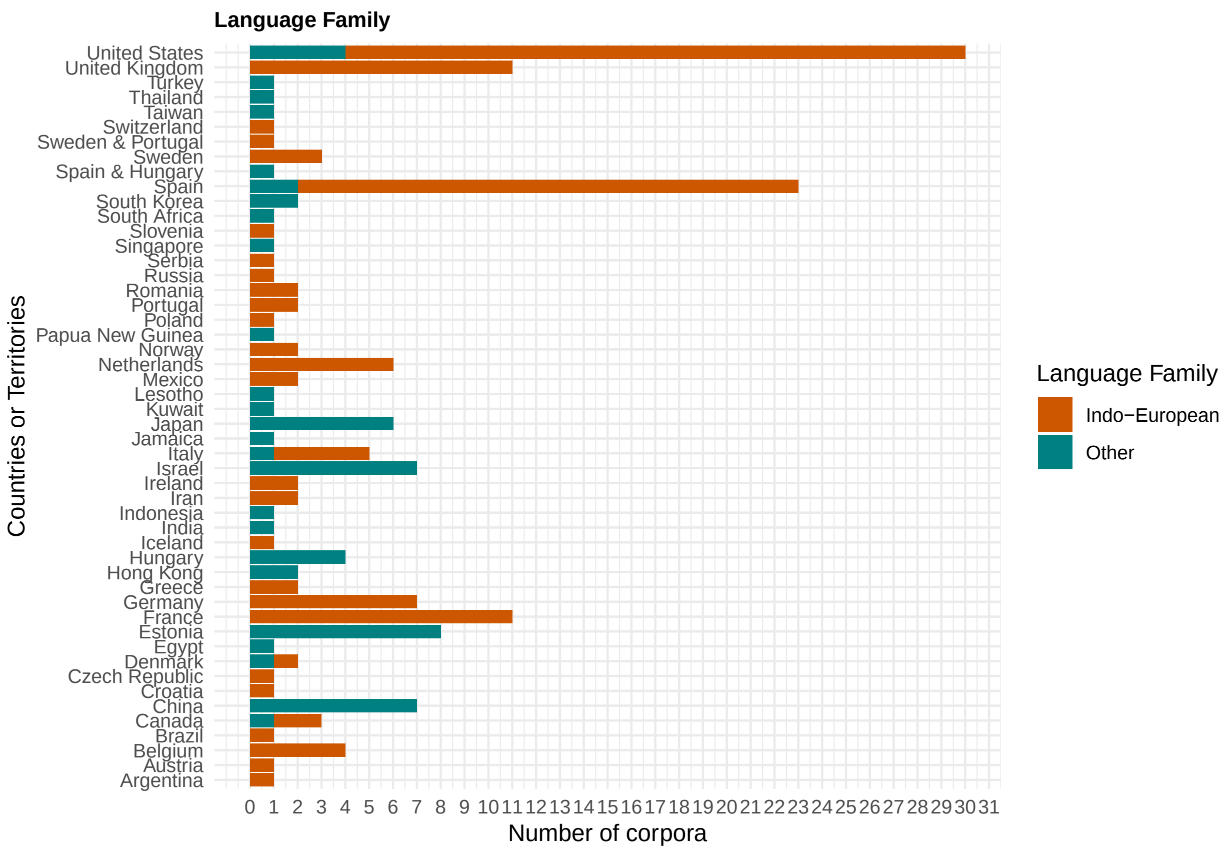

Figure 8. Distribution of Indo-European Languages by Country.

represent the world population, nor faithfully represent the countries in which data were collected, in terms of SES, urbanization, family structure, and linguistic dimensions.

In terms of SES, our macro-level analyses showed that represented countries had a higher percentage of individuals completing lower secondary school, and higher Gross Domestic Product (GDP) per capita, compared to the worldwide distribution. Our analyses further suggested that higher SES families with highly educated parents were more common than would have been expected given their source countries. Considering the extensive literature linking SES and language acquisition, these are important biases to take into account.

Notably, approximately half of the corpora feature parents engaged in research-related professions, with academics frequently studying their own (grand-)children.

In comparison, in the US general population, in 2020, only 6% of American citizens would fall under such definitions. The danger of extrapolating from such a sample goes beyond issues previously discussed in the context of convenience (homogeneous or heterogeneous) samples (Bornstein, Jager, & Putnick, 2013), with parents pursuing a career in academia (and likely in language-related areas) possibly being unusual in their linguistic behavior.

Regarding urbanization, 91% of corpora were urban, compared to 59% of the world population, with only 5 corpora exclusively rural. The latter seemed to be reasonable given urbanization statistics for the countries where data were collected. Thus, in this case, the bias pertains only to which countries are represented, and not how data are sampled within the country. One challenge for future work will involve teasing apart the ways in which urbanization may relate to language outcomes, including not only the parental goals mentioned in the Introduction (Keller, 2012) but also factors that may be confounded with urbanization. For example, a study using long-form recordings suggested that infants growing up in formerly rural families that had migrated to urban centers were afforded less spoken input, and had smaller vocabularies, than infants whose families had remained rural (Ma et al., 2024).

In terms of family structure, the represented countries had smaller households compared to the world average. When comparing household composition in our CHILDES sample with the countries where data had been collected, we found a possible trend for underrepresenting extended families. A bias towards nuclear families limits our understanding of diverse caregiving structures, particularly alloparental involvement, in language acquisition.

While multi-child families are well-represented in our CHILDES sample, the possible absence of other household members' speech in some transcripts (e.g., siblings' speech, Loukatou, Scaff, Demuth, Cristia, & Havron, 2022) could hinder insights into typical daily language interactions. Limited data on birth order also restricts robust conclusions on sibling configuration in CHILDES and underscores the need for broader exploration of non-parental caregiving roles typically overlooked in studies focusing on predominantly WEIRD households.

Finally, we found that English and more broadly Indo-European languages were overrepresented in our CHILDES sample, a point already made by others (e.g., Christiansen, Contreras Kallens, & Trecca, 2022). This aligns with trends in child development literature (Kidd & Garcia, 2022; Singh, Cristia, et al., 2023). Interestingly, out of the 10 countries with the most utterances (Figure 2), 5 were countries or territories where non-Indo-European languages are spoken: China, Israel, Japan, Indonesia, and Hong Kong. Additionally, although a third of the corpora were not monolingual, this nonetheless underrepresents multilingualism in the world's population (Grosjean, 2024).

These biases likely have both proximal and distal consequences for child language. Proximally, studies on typologically diverse languages (e.g., Casillas et al., 2024; De León, 1999; Demuth, 2022; C. L. Pye, 1980) highlight the limitations of current acquisition theories to the extent they rely on biased samples. For example, theories like syntactic bootstrapping, which assume consistent exposure to specific sentence structures, were developed in the context of Indo-European language. These can struggle to generalize to free word order languages such as Warlpiri, and tend to stress the syntactic rather than morphological side of bootstrapping, as English is relatively morphologically impoverished language (Babineau, Havron, Dautriche, Carvalho, & Christophe, 2023).

Broader socioecological diversity, such as variations in SES that influence familial stress (HART, SPERBER, & NOBLE, 2024), or polyadic caregiving environments that shape input sources and conversational dynamics (Sagna, Vihman, & Brown, 2022), may open new ways of thinking about the forces driving some of our classical findings on SES and input. Multilingual settings add another layer of complexity; for instance, Thai-English bilinguals demonstrate distinct narrative styles and gestural patterns depending on the language used, reflecting cultural norms and linguistic features (Rochanavibhata & Marian, 2022). More distal factors, such as noise pollution—more common in dense urban areas— linked with early brain development (Simon, Merz, He, & Noble, 2022) can have downstream effects on language acquisition. Skewed samples not only hinder the testing and generalization of language acquisition theories, but also risk mischaracterizing developmental mechanisms more broadly.

Although our CHILDES sample is more homogeneous than expected given global and national variations, there remains substantial diversity across all the dimensions we studied. However, these dimensions are also confounded in these CHILDES corpora. Put simply, since most samples are from nuclear families with well-educated parents living in urban sites, it is impossible to tease apart these three dimensions. Additionally, this confound makes it hard to understand the intersectionality of such factors.

While making some strides in documenting the demographic characteristics of the people represented in naturalistic corpora in CHILDES, our work has several limitations. First, we focused on SES, urbanization, family structure, and language, but other dimensions may well be relevant for language acquisition research. We hope that our sharing of materials will facilitate others' addition of novel descriptors. Second, the absence of comprehensive data for certain variables restricted a thorough analysis of representativeness on some dimensions. This should not be seen as the fault of the corpus curators, as these data were not always available or were not considered relevant at the time. It is only recently that researchers started sounding the alarm regarding underreporting of relevant demographic characteristics (e.g., Singh, Barokova, et al., 2023; Singh, Cristia, et al., 2023).

Our paper emphasizes the importance of evaluating the diversity of the samples we rely on. This issue is not unique to CHILDES, and applies broadly to any developmental archive (e.g., Databrary, Gilmore, Adolph, & Millman, 2016). In the era of big data, AI, and the rise of child-focused technologies, it is essential to consider the generalizations that models based on such data might produce for language acquisition - and reflect on how these models and generalizations impact the entire developmental field.

Researchers using CHILDES data should familiarize themselves with the demographic characteristics of their samples and acknowledge any limitations that could affect their findings. For future contributions, we recommend including demographic metadata following CHILDES's already existing metadata system. Requiring data entry would ensure the inclusion of key descriptors at both the speaker and corpus levels, improving the database's demographic information. When collecting data, purposive sampling can help include all key sub-groups for a representative sample (see Bornstein et al., 2013). Collaborating with local communities and adopting participatory practices (Gehlert & Mozersky, 2018) can enhance data collection in diverse contexts.

Finally, systemic changes in academia are crucial for fostering long-term diversity. Journals, editorial boards, and funding agencies should create opportunities and provide support for studies on underrepresented populations and researchers. Adopting key performance indicators for diversity—focusing on linguistic, geographic, and socioeconomic representation (Havron, Scaff, Hitczenko, & Cristia, 2022)—and increasing the inclusion of non-Western authors in developmental science (Aravena-Bravo et al., 2024) are actionable steps to address systemic biases.

To conclude, we found that many countries and languages are represented in CHILDES, but that these countries did not constitute a representative sample of the world's countries, and that for many dimensions, families were not representative of their own countries. We also noted low variability along certain dimensions (e.g., SES and family structure) and the predominance of Indo-European languages, with English having the most recorded corpora. Despite these limitations, CHILDES remains an unparalleled resource that has contributed significantly to the field over decades, exemplifying the success of open and collaborative science. Our discussion leads us to argue for the systematic inclusion of descriptors in speaker-level and corpus-level CHILDES metadata, to track and quantify both diversity and representativity.

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